A

Mini Project Report

on

MEDITOUCH - Smart Health Reminder

Submitted in partial fulfillment of the requirements for the

degree

Second Year Engineering – Computer Science Engineering (Data Science)

by

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Academic year: 2025-26

CERTIFICATE

This to certify that the Mini Project report on **Meditouch** has been submitted by Krrish Soni (24107066), Udayraj Vishwakarma (24107138), Shravani Tatkare (24107121) and Anushka Pokharkar (24107013) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

Prof. Priyanka Kahade
Guide

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Principal

External Examiner:
1.

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Place: A. P. Shah Institute of Technology, Thane
Date:

ACKNOWLEDGEMENT

This project would not have come to fruition without the invaluable help of our guide **Priyanka Kahade**. Expressing gratitude towards our HoD, **Dr. Pravin Adivarekar**, and the Department of Computer Science Engineering (Data Science) for providing us with the opportunity as well as the support required to pursue this project. We would also like to thank our project coordinator **Prof. Richa Singh and Prof Kiran Gaikwad** who gave us her valuable suggestions and ideas when we were in need of them. We would also like to thank our peers for their helpful suggestions.

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Chapter 1: Introduction

In the contemporary healthcare landscape, the integration of digital technologies has become essential for effective health management. With increasing chronic disease prevalence and the growing need for personalized healthcare solutions, mobile health applications have emerged as powerful tools for bridging the gap between patients and healthcare providers. MEDITOUCH addresses this need by providing a comprehensive, cross-platform health management solution that serves both patients and doctors through a unified interface.

The application leverages modern technologies including Flutter for cross-platform development, Firebase for real-time data synchronization, and Riverpod for state management. By combining medication tracking, appointment management, health vitals monitoring, and secure doctor-patient communication in a single platform, MEDITOUCH eliminates the fragmentation commonly found in existing health applications.

1.1 Purpose:

This document serves as the Mini Project Report for MEDITOUCH - Smart Health Reminder, developed as part of the Second Year Engineering curriculum in Computer Science Engineering (Data Science) at A.P. Shah Institute of Technology. The report provides a comprehensive overview of the project's objectives, design, implementation, and outcomes.

The MEDITOUCH application is designed to be a digital health guardian that assists patients in managing their medications, tracking health vitals, booking appointments with doctors, and maintaining seamless communication with healthcare providers. For doctors, the platform offers efficient appointment management, digital prescription capabilities, and real-time patient interaction tools.

1.2 Problem Statement

Despite the proliferation of health applications, several critical challenges persist in the digital health ecosystem. Medication non-adherence remains a significant issue, with studies indicating that approximately 50% of patients do not take their medications as prescribed. This non-compliance leads to poor health outcomes, increased hospitalizations, and elevated healthcare costs.

Existing health applications are predominantly fragmented, requiring users to install multiple apps for different health-related tasks. Furthermore, most solutions focus exclusively on either patient or doctor needs, lacking the bidirectional communication essential for effective healthcare delivery. There is no unified cross-platform solution that serves both patients and doctors with real-time synchronization capabilities.

1.3 Objectives

The primary objectives of the MEDITOUCH project are:

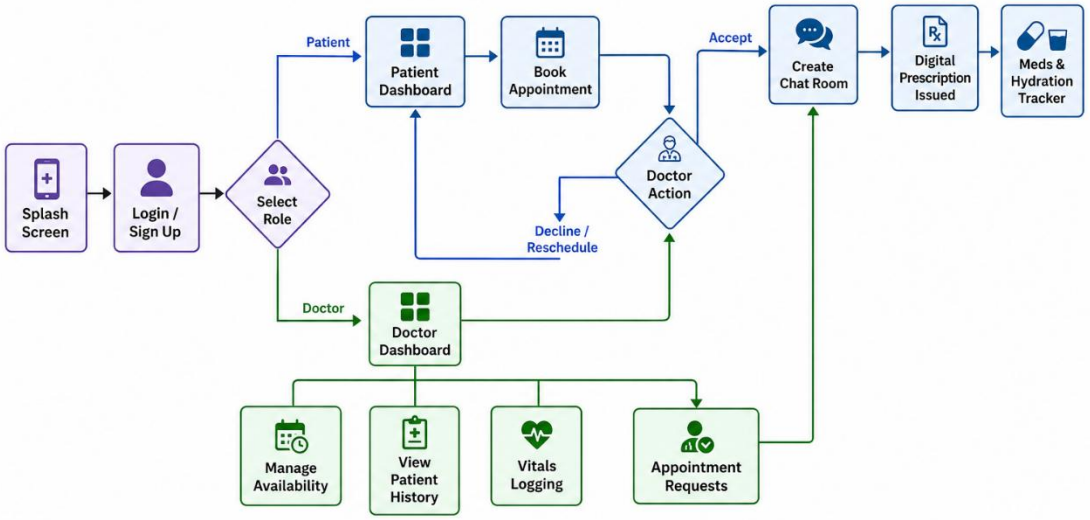
- **Develop a cross-platform health management application using Flutter framework.**
- **Implement intelligent medicine tracking with alarm-style reminder notifications.**
- **Enable real-time appointment booking and management between patients and doctors Provide comprehensive health vitals and hydration logging capabilities.**
- **Build an over-the-counter (OTC) symptom checker with medicine suggestions.**
- **Create a secure real-time chat system for doctor-patient communication Implement digital prescription management with dynamic medicine lists.**
- **Ensure real-time data synchronization using Firebase backend services.**

1.4 Scope

The MEDITOUCH application includes support for Android, iOS, Web, and Windows platforms. The system accommodates both patient and doctor user roles with role-based routing and feature access. Real-time features include appointment management, chat functionality, and notification delivery. The application implements both local notifications for medicine reminders and push notifications for system updates. The user interface follows a futuristic glassmorphic dark theme design.

Chapter 2: Proposed System

MEDITOUCH is a comprehensive health management platform designed to streamline the healthcare experience for both patients and medical practitioners. The system architecture follows a Repository-Service pattern with clear separation of concerns, enabling maintainable and scalable code organization.



2.1 Proposed system

This figure represents the workflow of a smart healthcare or telemedicine application. It starts with the splash screen, followed by login or sign-up, after which the user selects their role as either a patient or a doctor. If the user chooses the patient role, they are directed to the patient dashboard where they can book an appointment. The appointment request is then sent to the doctor, who can accept, decline, or reschedule it. If the doctor accepts the request, a chat room is created for communication, and the doctor can issue a digital prescription. Based on the prescription, the system also provides features like a medication and hydration tracker for the patient. On the other hand, if the user selects the doctor role, they are taken to the doctor dashboard where they can manage availability, view patient history, log patient vitals, and handle appointment requests. Overall, the diagram explains how both patients and doctors interact within the system, showing the complete flow from login to consultation and post-treatment tracking.

In the doctor path (usually shown in green), the doctor accesses their dashboard and can manage availability, view patient history, log vitals, and handle appointment requests. The horizontal layout highlights parallel functionality—showing both roles working simultaneously—while arrows clearly indicate the direction of data flow and decision points. Overall, this design improves readability, visually separates responsibilities, and demonstrates the complete lifecycle of a consultatio

Chapter 3: Project Outcomes

The MEDITOUCH project has successfully achieved the following outcomes:

- Developed a fully functional cross-platform health management application supporting Android, iOS, Web, and Windows.
- Implemented a comprehensive medicine tracking system with local alarm-style notifications and adherence monitoring.
- Created a real-time appointment booking and management system with bidirectional status updates.
- Achieved real-time data synchronization across all user interactions using Firebase Firestore.
- Implemented a modern glassmorphic UI design with animated backgrounds and intuitive navigation.
- Built a secure chat system enabling seamless doctor-patient communication. Developed digital prescription functionality with dynamic medicine list management.
- Learning outcomes include proficiency in Flutter framework development, Firebase backend integration, real-time database design, and cross- platform mobile application architecture.
- The project also enhanced understanding of healthcare domain requirements and user experience design principles.

Chapter 4: Software Requirements

Component	Minimum Requirement
Processor	Intel Core i3 or AMD equivalent
RAM	4 GB minimum, 8 GB recommended
Storage	10 GB free disk space
Display	1366 x 768 resolution
Internet	Broadband connection for Firebase sync

Table 4.1: Hardware Requirements

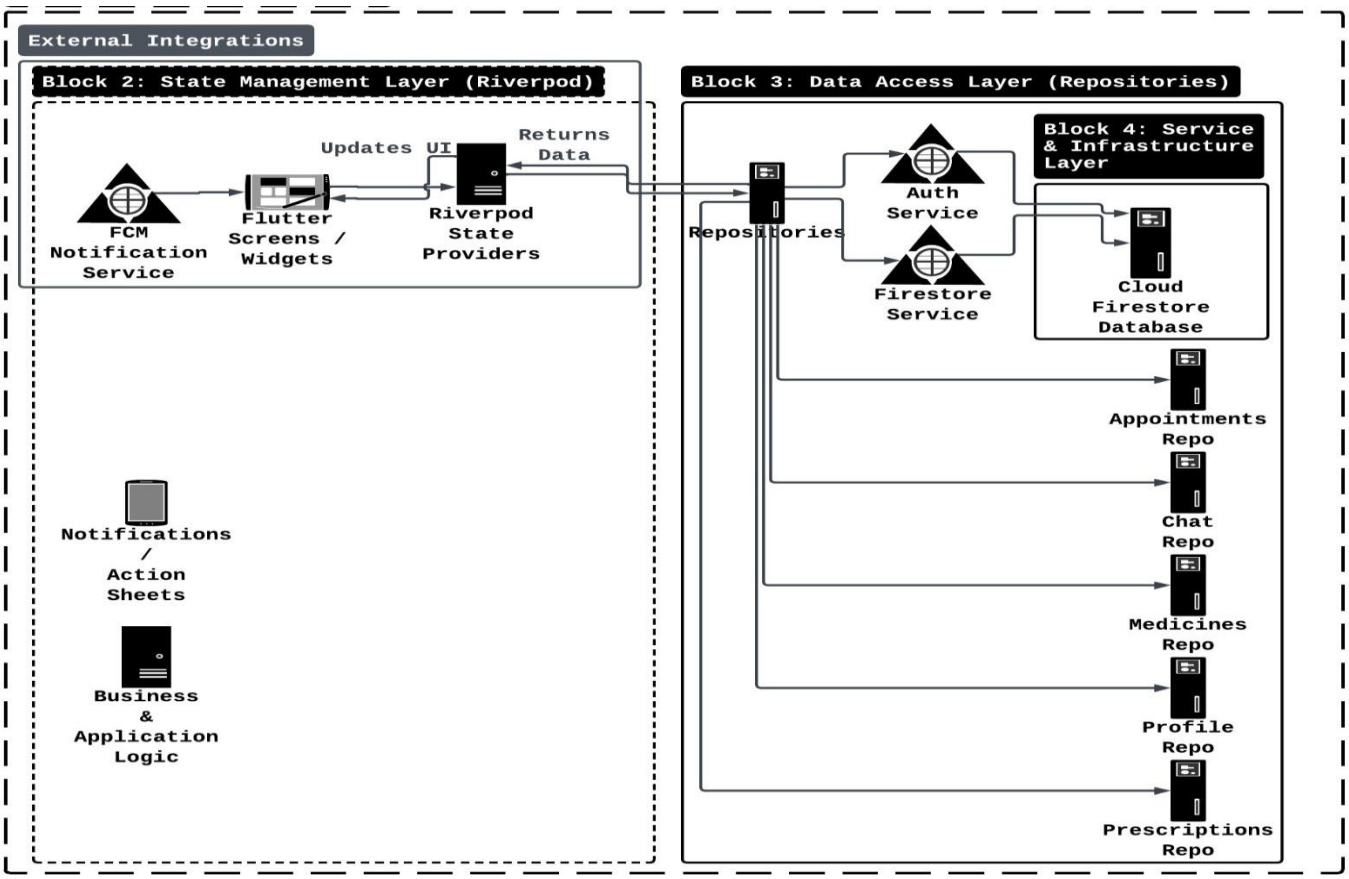
Software	Version/Details
Flutter SDK	3.24.0 or higher
Dart	3.5.0 or higher
Android Studio	Latest stable version
Firebase	Firebase CLI and console access
Git	2.30 or higher
Chrome	For web debugging

Table 4.2: Software Requirements

Technology	Purpose
Flutter	Cross-platform UI framework
Riverpod	State management
Firebase Firestore	Real-time NoSQL database
Firebase Authentication	Email/password authentication
Firebase Cloud Messaging	Push notifications
flutter_local_notifications	Alarm-style local reminders
Google Fonts	Typography (Poppins, Inter)

Table 4.3: Technology stack

Chapter 5: Project Design



5.1 Project design

This diagram represents a **4-block architecture** of your healthcare app, showing how different layers of the system are organized and how they interact with each other.

In simple terms, the architecture is divided into four main layers. The **State Management Layer (Block 2)** handles the app's UI and logic using Flutter and Riverpod. The Flutter screens (UI) display data to the user and send user actions (like booking appointments). Riverpod state providers manage this data flow by updating the UI and returning updated data. This layer also connects with external services like Firebase Cloud Messaging (FCM) for notifications, which are shown to users through notifications or action sheets. It basically acts as the bridge between the user interface and the backend logic.

Next is the **Data Access Layer (Block 3)**, which uses repositories. These repositories act like middlemen between the app and backend services. They fetch and send data without exposing the UI directly to the database or services. For example, there are separate repositories for appointments, chat, medicines, profile, and prescriptions. This keeps the code clean and organized.

Then comes the Service & Infrastructure Layer (Block 4), which includes core backend services like authentication and database. The authentication service handles user login and security, while the Firestore service connects to the cloud database where all app data is stored. These services interact with the repositories to process and store data.

Chapter 6: Project Scheduling

GANTT CHART

Smartsheet Tip → A Gantt chart's visual timeline allows you to see details about each task as well as project dependencies.

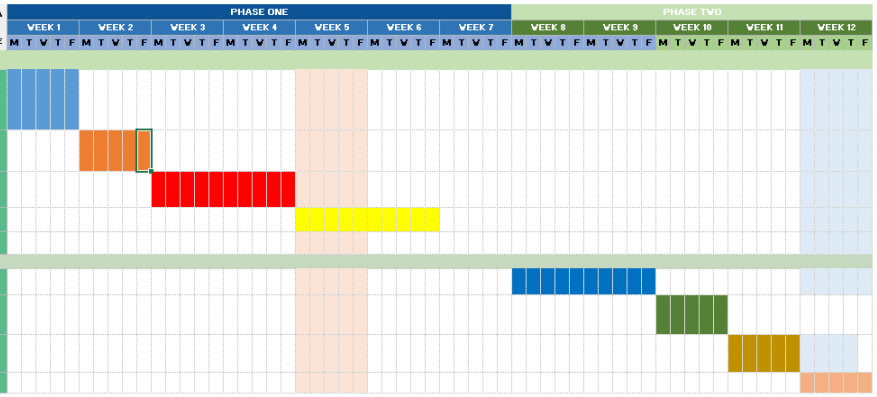
PROJECT TITLE: MEDITOUCH

PROJECT GUIDE: Prof. Priyanka Kahade

INSTUTUTE & DEPARTMENT NAME AP SHAH INSTITUTE OF TECHNOLOGY(CSE-Data Science)

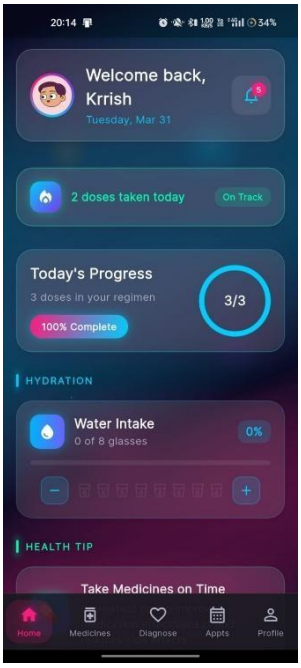
DATE: 13/4/26

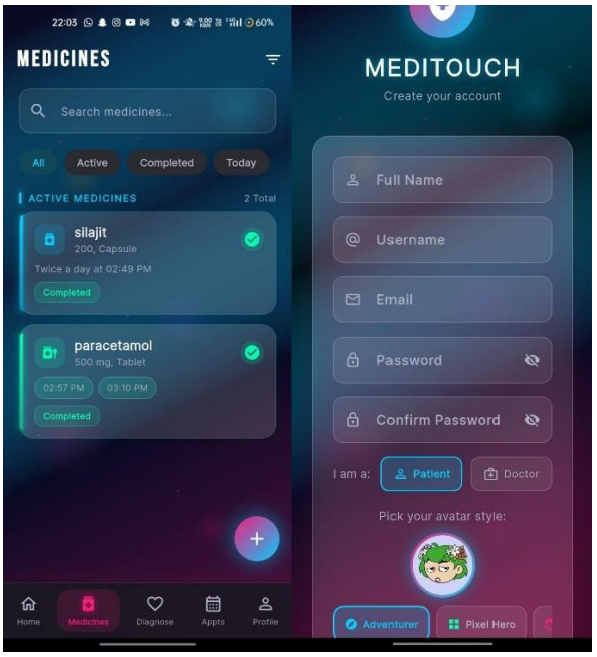
WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION(Weeks)	PERCENTAGE OF TASK COMPLETE
1	**Project Conception and Initiation**					
1.1	Group formation and Topic finalization. Identifying the scope and objectives of the Mini Project	All	1-13-26	1-16-26	1	100%
1.2	Identifying the functionalities of the Mini Project	All	1-16-26	1-20-26	1	100%
1.3	Discussing the project topic with the help of paper prototype.	All	1-20-26	2-3-26	2	100%
1.4	Designing the Graphical User Interface(GUI)	Kirish Soni, Udayraj vishwakarma	2-3-26	2-17-26	2	100%
1.5	Presentation I	All	2-17-26	2-24-26	1	100%
2	**Project Design and Implementation**					
2.1	Database Design	Yash Shinde and Swati Vishwakarma	2-24-26	3-10-26	2	100%
2.2	Database Connectivity of all modules	Shravani Tatakare, Udayraj vishwakarma	3-10-26	3-17-26	1	100%
2.3	Integration of all modules and Report Writing	Anushka, Udayraj Vishwakarma	3-17-26	3-24-26	1	100%
2.4	Presentation II	All	3-24-26	4-1-26	1	100%

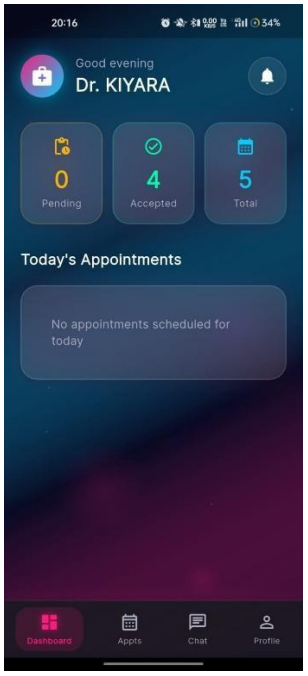


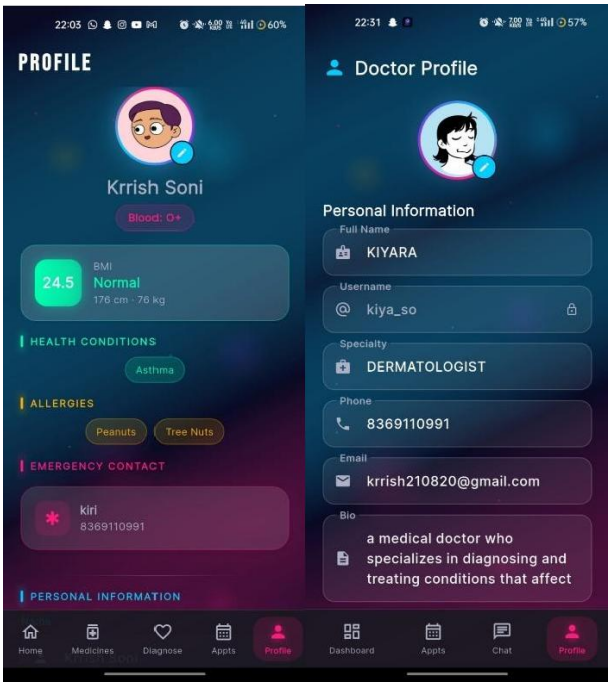
Smart Health Reminder Project

Chapter 7: Results









Chapter 8: Conclusion

MEDITOUCH - Smart Health Reminder represents a comprehensive solution to the fragmented digital health ecosystem. By integrating medication tracking, appointment management, health vitals monitoring, and secure doctor-patient communication into a single cross-platform application, the project successfully addresses the key challenges identified in the problem statement. The dual-role architecture ensures that both patients and healthcare providers benefit from streamlined workflows and real-time data synchronization.

All eight project objectives have been successfully achieved. The Flutter-based cross- platform application delivers a consistent user experience across Android, iOS, Web, and Windows platforms. The medicine tracking system with alarm-style notifications helps improve medication adherence, while the real-time appointment and chat systems facilitate effective doctor-patient communication. The digital prescription feature streamlines the medication management process for healthcare providers.

Future Enhancements

AI-powered health predictions: Implement machine learning algorithms to analyze user health data and provide personalized health recommendations and risk assessments.
Wearable device integration: Connect with fitness trackers and smartwatches to automatically sync health vitals and activity data. Video consultation: Add secure video calling capabilities for remote consultations between patients and doctors. Multi-language support: Expand accessibility by supporting multiple regional languages for broader user adoption.Health report PDF export: Enable users to generate and download comprehensive health reports in PDF format for offline sharing and record-keeping.

References

Flutter Documentation, Google. Available at: https://flutter.dev/docs

Firebase Documentation, Google. Available at: https://firebase.google.com/docs

Riverpod Documentation. Available at: https://riverpod.dev

World Health Organization, "Adherence to Long-term Therapies: Evidence for Action," WHO, 2003.

A. Kumar et al., "Mobile Health Technology: A Novel Tool in Chronic Disease Management," International Journal of Medical Informatics, vol. 145, 2021.

S. Patel and R. Sharma, "Smart Medication Reminder Systems: A Comprehensive Review," Journal of Healthcare Engineering, vol. 2020, 2020.

M. Smith et al., "The Impact of Mobile Health Applications on Patient Engagement and Outcomes," Digital Health Journal, vol. 7, pp. 1-15, 2021.